

```

clc; clear; close all;
Users = 1:8;
TotalSlots = 5;
Alloc = zeros(1,8);
Alloc(1:TotalSlots) = 1;
Blocked = 1 - Alloc;
disp('User Wise Slot Allocation')

```

User Wise Slot Allocation

```

disp(table(Users',Alloc', 'VariableNames',
{'User', 'Status(1=Allocated,0=Blocked)'}))

```

| User | Status(1=Allocated,0=Blocked) |
|------|-------------------------------|
| 1    | 1                             |
| 2    | 1                             |
| 3    | 1                             |
| 4    | 1                             |
| 5    | 1                             |
| 6    | 0                             |
| 7    | 0                             |
| 8    | 0                             |

```

figure
subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on

subplot(2,1,2)
plot(Users,Alloc,'-o','LineWidth',2)
hold on
plot(Users,Blocked,'-s','LineWidth',2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on

```

